gGA Dimer Scans: U/t Dependence at $\Delta V = 0$ and $\Delta V = 2$

(Ghost Multiplicities $N_g = 2$ and $N_g = 4$, Direct-Minimization Solver)

gGut_mau project — `not_imp_gga_dimer.py`

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Scope. This note collects the U/t scans produced by the direct-minimization ("not_imp") gGA solver for the half-filled Hubbard dimer, at zero and finite site-splitting potential ΔV (see `not_impurity.tex` for the solver's equations and `code.tex` for the ΔV generalization). Four scans are collected:

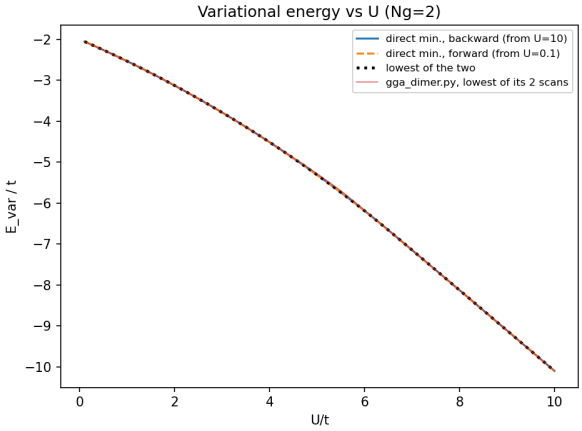
- `scan_ng2_not_imp` — $N_g = 2$, $\Delta V = 0$;
- `scan_ng4_not_imp` — $N_g = 4$, $\Delta V = 0$;
- `scan_ng2_dV2_confirmed` — $N_g = 2$, $\Delta V = 2$, cross-checked against the impurity-problem solver;
- `scan_ng4_dV2_confirmed` — $N_g = 4$, $\Delta V = 2$, cross-checked against the impurity-problem solver.

All scans sweep $U/t \in [0.1, 10.0]$ in steps of 0.1, at $t = 1$. The "confirmed" scans report the lowest-energy solution after reconciling the two independently implemented solvers (impurity-problem and direct-minimization); see commit `fe06bd6`.

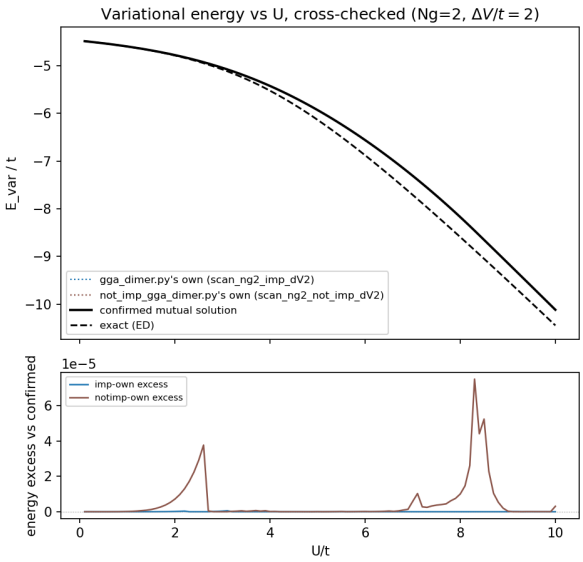
1 $N_g = 2$

1.1 Total energy vs. U/t

Figure 1 shows the variational ground-state energy E_{var}/t as a function of U/t, at $\Delta V = 0$ and $\Delta V = 2$.



(a) $\Delta V = 0$

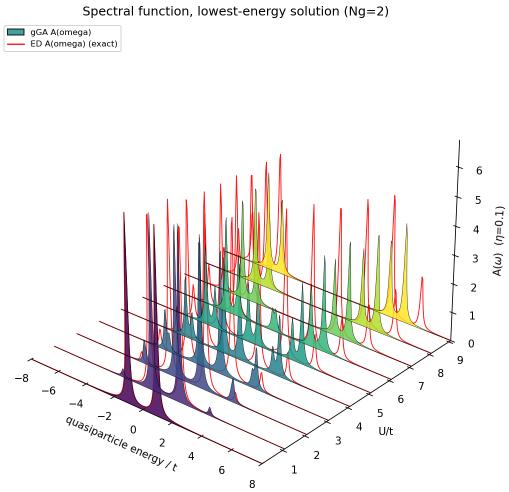


(b) $\Delta V = 2$

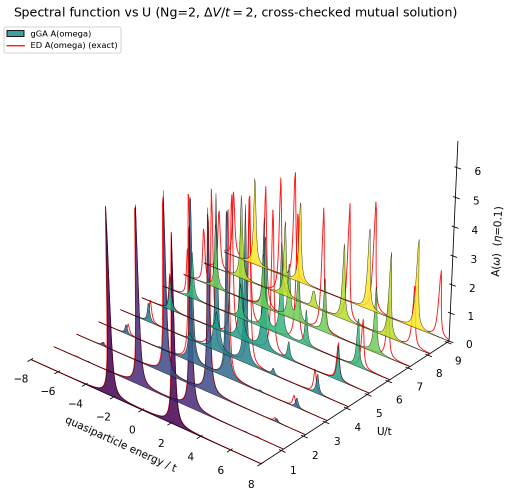
Figure 1: $N_g = 2$: total energy E_{var}/t vs. U/t.

1.2 Quasiparticle spectrum: 3D view

Figure 2 shows the full quasiparticle energy spectrum $\{\epsilon_n^{qp}\}$ as a function of U/t (ridgeline / 3D representation).



(a) $\Delta V = 0$



(b) $\Delta V = 2$

Figure 2: $N_g = 2$: quasiparticle spectrum vs. U/t, 3D view.

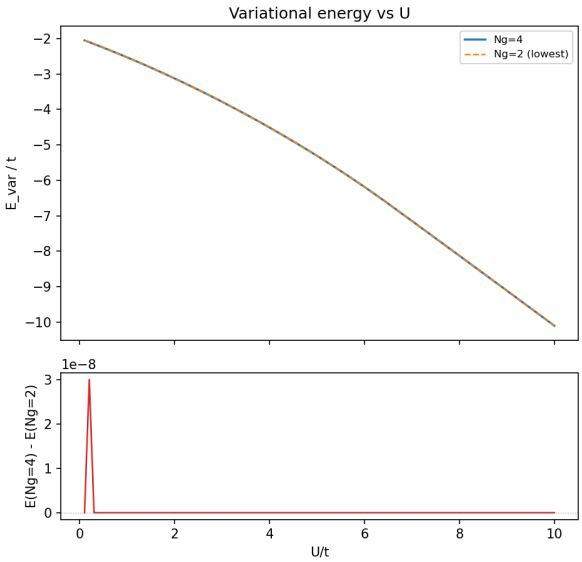
1.3 Spectral function movies

- $\Delta V = 0$: `scan_ng2_not_imp/A_omega_vs_U.mp4`
- $\Delta V = 2$: `scan_ng2_dV2_confirmed/A_omega_vs_U.mp4`

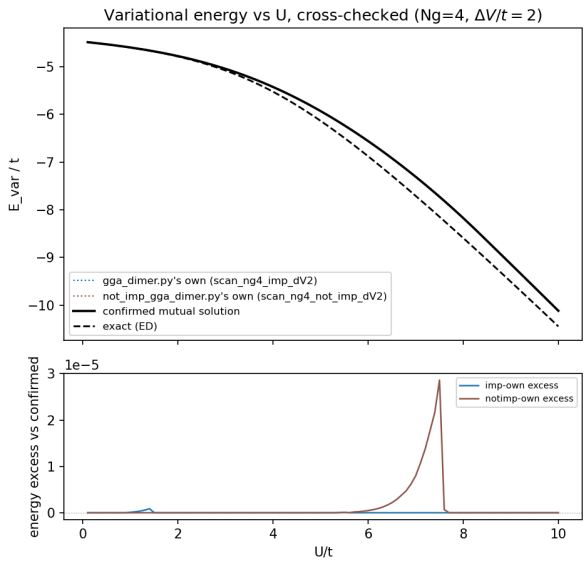
(Links open the file with the PDF viewer's associated application; if the viewer does not support `run:` links, open the path directly.)

2 $N_g = 4$

2.1 Total energy vs. U/t



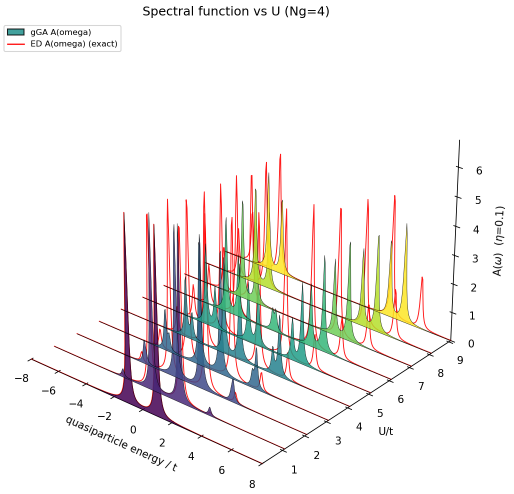
(a) $\Delta V = 0$



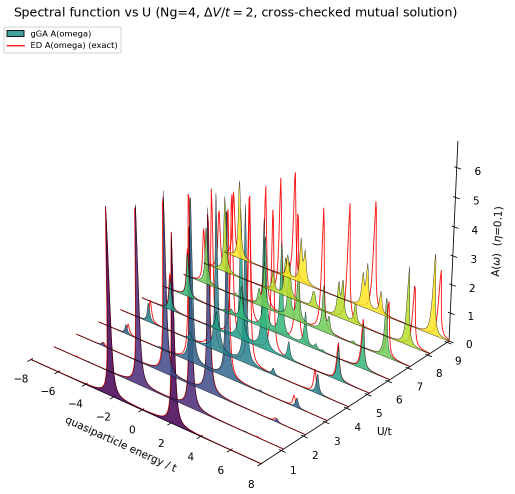
(b) $\Delta V = 2$

Figure 3: $N_g = 4$: total energy E_{var}/t vs. U/t. The $\Delta V = 0$ panel also reports $E(N_g = 4) - E(N_g = 2)$ (see Section 3).

2.2 Quasiparticle spectrum: 3D view



(a) $\Delta V = 0$



(b) $\Delta V = 2$

Figure 4: $N_g = 4$: quasiparticle spectrum vs. U/t, 3D view. The larger ghost multiplicity doubles the number of quasiparticle branches compared to Figure 2.

2.3 Spectral function movies

- $\Delta V = 0$: `scan_ng4_not_imp/A_omega_vs_U.mp4`
- $\Delta V = 2$: `scan_ng4_dV2_confirmed/A_omega_vs_U.mp4`

3 Discussion: $\Delta V = 0$ vs. $\Delta V = 2$

Weak coupling: ΔV dominates. At small U/t the two scans differ substantially. At $U/t = 0.1$, $E_{\text{var}}/t = -2.0503$ for $\Delta V = 0$ but -4.4823 for $\Delta V = 2$ (both N_g). This is the expected $U \to 0$ limit: with $\Delta V = 0$ the dimer's noninteracting energy is $-2t = -2$ plus a small $O(U)$ correction, while at $\Delta V = 2$ the site-splitting opens a one-body level gap and the $U \to 0$ energy approaches the closed form $E = -2\sqrt{t^2 + \Delta V^2} = -2\sqrt{5} \approx -4.472$ (see `project_deltaV_gga` validation). The gap between the two curves is largest here, $\Delta E \approx 2.43$, because ΔV acts as an unscreened one-body potential that the (still weak) correlations barely renormalize.

Strong coupling: the curves converge. As U/t grows the two energies draw close together: at $U/t = 10$, $E_{\text{var}}/t = -10.101$ ($\Delta V = 0$) vs. -10.120 ($\Delta V = 2$), a difference of only ≈ 0.02 – two orders of magnitude smaller than at weak coupling. In the Mott-like regime the total energy is governed almost entirely by U (both sites approach single occupancy), and the fixed $\Delta V = 2$ site-splitting becomes a small perturbation on top of the dominant Hubbard physics. The quasiparticle spectra (3D plots, Figs. 2, 4) show the same trend: the $\Delta V = 2$ branches, well separated at small U, bunch together with the $\Delta V = 0$-like structure as U/t increases.

Branch structure. At $\Delta V = 0$ the lowest-energy solution alternates between "forward" and "backward"-seeded branches as U/t is swept (see `scan_ng2_not_imp/lowest_energy.txt`), reflecting a near degeneracy between the two sweep directions that the scan resolves by keeping the

lower of the two at each U. At $\Delta V = 2$ this degeneracy is lifted by the site-splitting itself, and the "confirmed" scans instead report a single branch cross-checked between the impurity-problem and direct-minimization solvers. The $\Delta V = 2$ quasiparticle energies show a sharp rearrangement between $U/t = 0.6$ and $U/t = 0.7$ (e.g. $N_g = 2$: $\epsilon^{qp} \approx -2.00 \to -3.12$), visible as a kink in the 3D plots; this is the level crossing where the ΔV-induced one-body gap stops being the dominant scale and the Hubbard repulsion takes over.

Ghost multiplicity ($N_g = 2$ vs. $N_g = 4$). At both $\Delta V = 0$ and $\Delta V = 2$, doubling the ghost multiplicity from $N_g = 2$ to $N_g = 4$ leaves the lowest variational energy numerically unchanged over the whole scanned range: `scan_ng4_not_imp/summary.txt` reports $E(N_g{=}4) - E(N_g{=}2) = 0$ to the printed precision at essentially every U/t (largest deviation $\sim 3 \times 10^{-8}$, at the level of solver tolerance), and comparing `scan_ng2_dV2_confirmed/lowest_energy.txt` against `scan_ng4_dV2_confirmed/lowest_energy.txt` shows the same agreement at $\Delta V = 2$ (e.g. $U/t = 1.0$: -4.593786 vs. -4.593786). The extra ghost degrees of freedom introduced at $N_g = 4$ therefore stay energetically inert for this dimer at both values of ΔV in this U range – consistent with the known $N_g \geq 2$ "sleeping ghost" caveat (`project_deltaV_gga`): the additional branches exist in the quasiparticle spectrum (compare Figs. 2 and 4, which show twice as many ϵ_n^{qp} branches at $N_g = 4$) but do not lower the ground-state energy below what $N_g = 2$ already finds.